

VIVIAN (JINGCHENG) YU

Department of Statistics and Actuarial Science, University of Waterloo
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EDUCATION

Ph.D. in Actuarial Science, University of Waterloo Sep 2024 – present

Advisor: Ruodu Wang. GPA: 91%.

Master of Financial Engineering, University of California, Los Angeles Dec 2022

UCLA Anderson School of Management. GPA: 3.82/4.0.

B.S. in Actuarial Mathematics / Management, University of Manitoba Dec 2019

First Class Honours; Dean's Honour List. GPA: 4.13/4.50 (major 4.21/4.50).

HONORS & AWARDS

- James C. Hickman Scholar, Society of Actuaries
- First Class Honours and Dean's Honour List, University of Manitoba, 2019

PROFESSIONAL CREDENTIALS

- Fellow of the Society of Actuaries (FSA)
- Chartered Enterprise Risk Analyst (CERA)
- CFA Level II Candidate

PREPRINTS

Wang, R. and Yu, J. (2026). Submodular risk measures. arXiv:2603.01232.

WORKING PAPERS AND WORK IN PROGRESS

Wang, R. and Yu, J. ES-only e-backtesting for AR-GARCH losses. Working paper, 2026.

Wang, R., Xia, C. and Yu, J. Mathematics of scenario-based risk evaluation. Work in progress, 2026.

TEACHING

Teaching Assistant, University of Waterloo Sep 2024 – present

Tutorials and mentorship for undergraduate actuarial science courses.

Teaching Assistant, University of California, Los Angeles Sep – Dec 2022

Tutorial lectures on stochastic calculus; grading and exam proctoring.

Teaching certifications: Fundamentals of University Teaching Program (completed); Certificate in University Teaching (in progress), University of Waterloo.

PROFESSIONAL EXPERIENCE

Actuarial Consultant, Ernst & Young, New York Jan – Sep 2023

Portfolio optimization with liabilities and reinvestments via simulated annealing and genetic algorithms in Python; reduced financing cost by 50%.

Actuarial Student Assistant, California Department of Insurance Jun – Sep 2022

Research on ISO filings and guideline revisions; redeveloped the Class Plan Application in VBA, reducing processing time by 30%.

PROFESSIONAL SERVICE

Reviewer and Assessment Grader, Society of Actuaries Aug 2022 – Mar 2024

Review Committee member for research paper evaluations (e.g., “Big Tech & Insurance”, “Zero Knowledge Proof”); end-of-module assessment grader.

OTHER RESEARCH PROJECTS

Classification and decision tree models for portfolio design Mar – Dec 2022

MFE capstone with UCLA Anderson and Chantico Global: CART models across 14 asset classes and 73 macroeconomic variables with rolling-window analysis.

Triggered shorting strategy on VIX May – Jun 2022

IAQF academic competition: regime identification via hidden Markov and clustering models feeding neural network predictions.

SKILLS

Programming: Python, R (Tidyverse), SQL, TensorFlow, VBA, $\text{\LaTeX}$. Languages: English, Mandarin Chinese.